OpenFlowでの サービスチェイニングによる SDNの実現

MPLS Japan 2013

株式会社エーアイコーポレーション

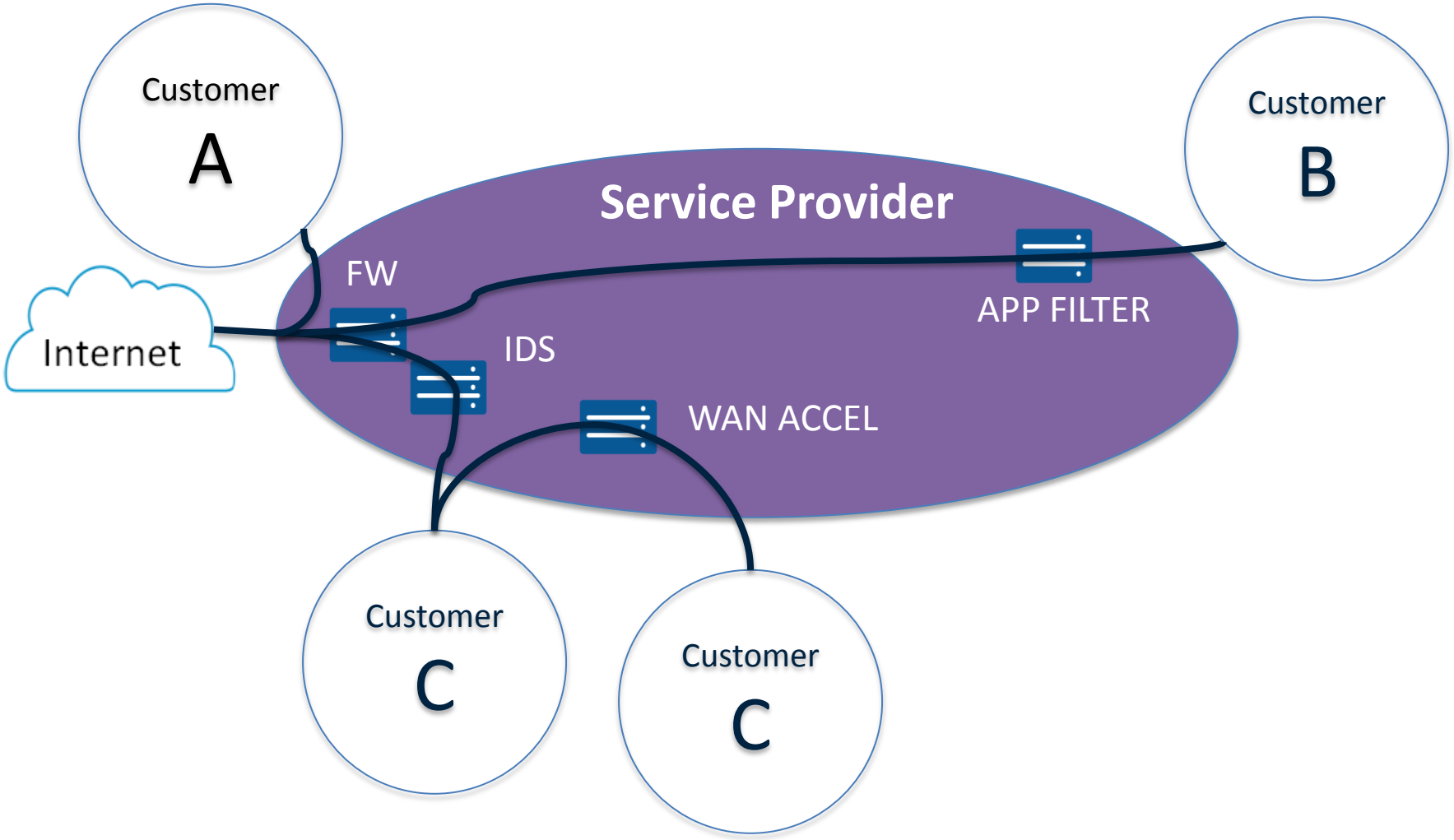
岩下 雅幸 / 溝口 茂

http://www.aicp.co.jp/

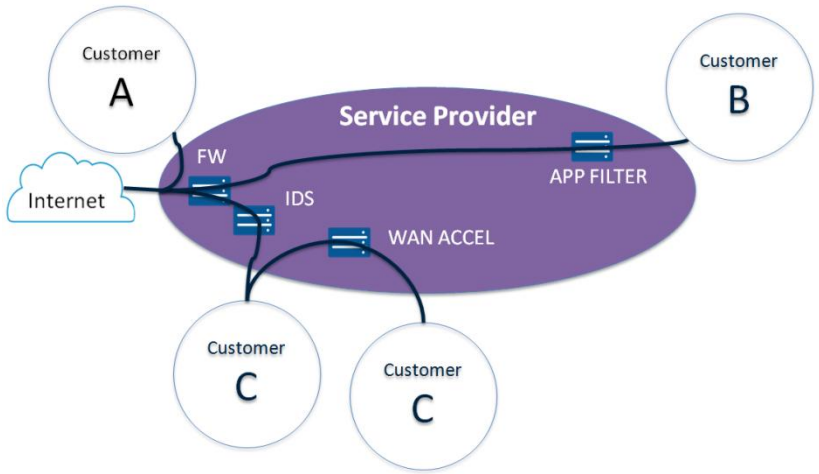
アジェンダ

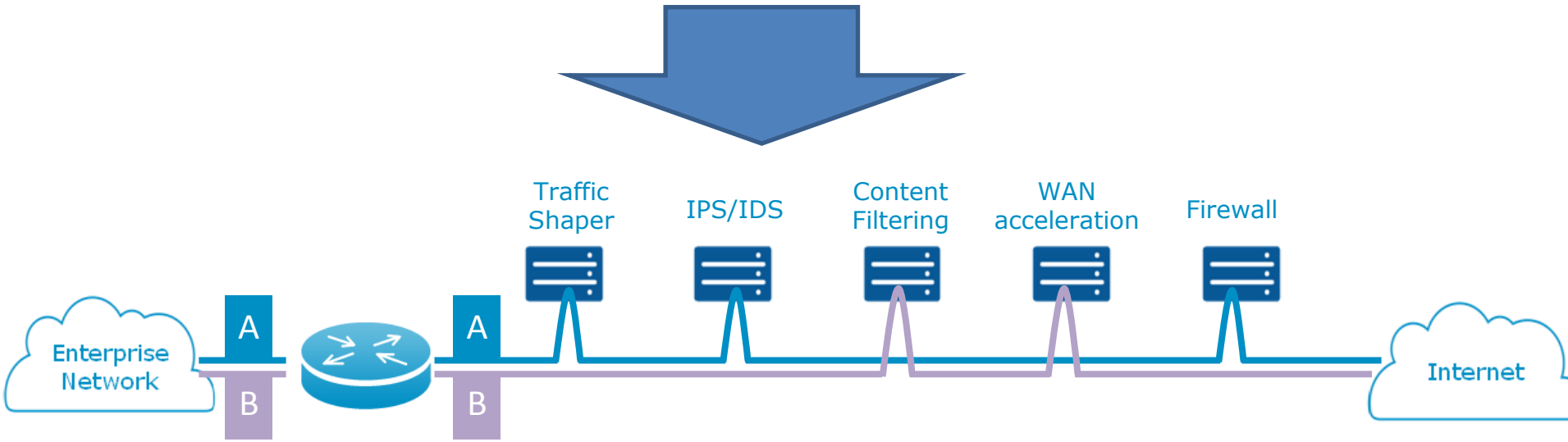
- Service Chaining とは？
- Service Chainingで要求されること
- Service Chainingのモデル化
- デモ
- 標準技術による汎用化

Service Chaining とは？

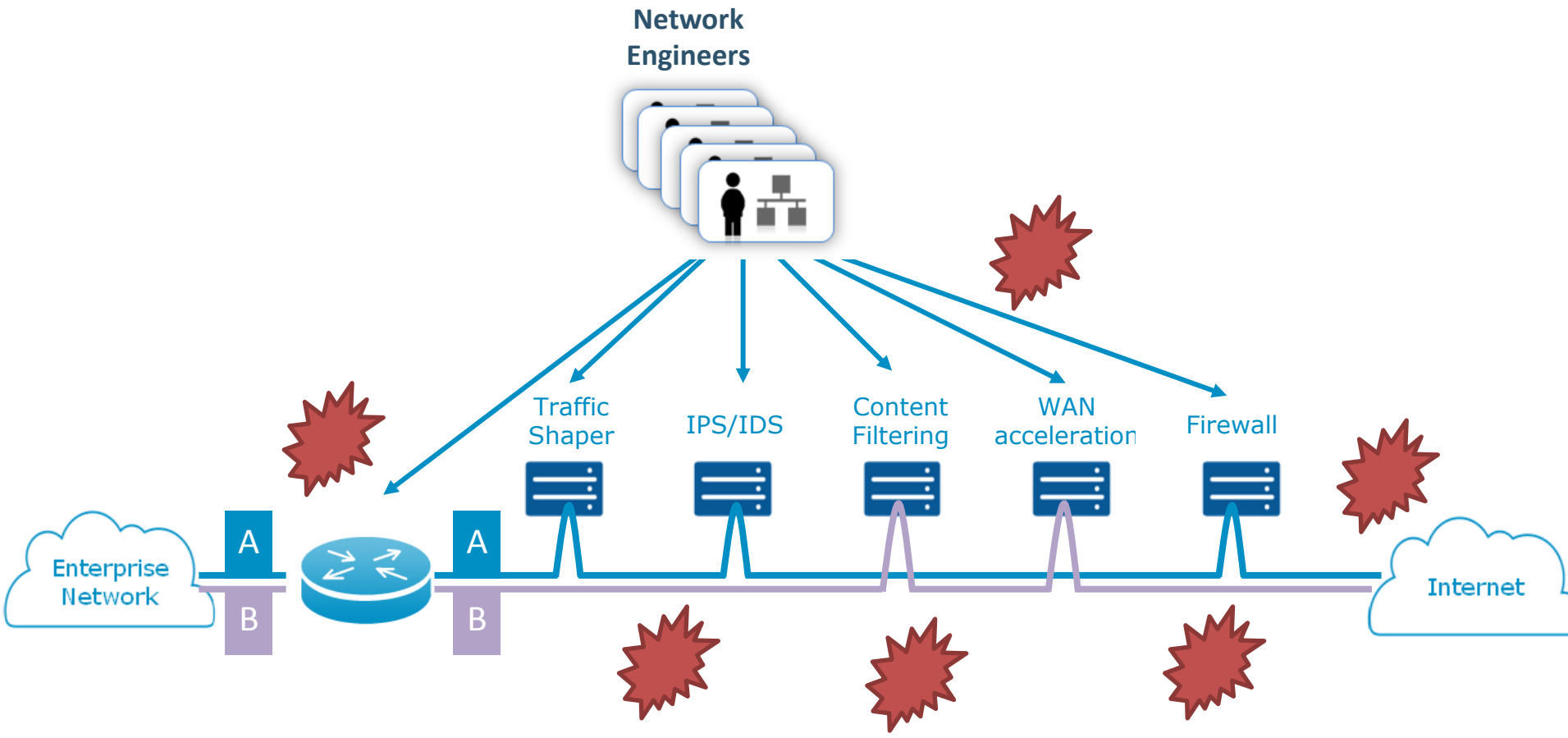


Service Chaining のイメージ

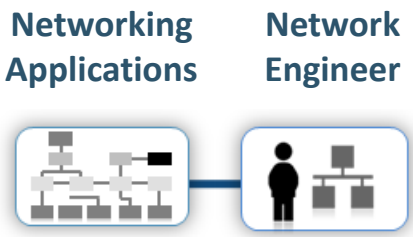


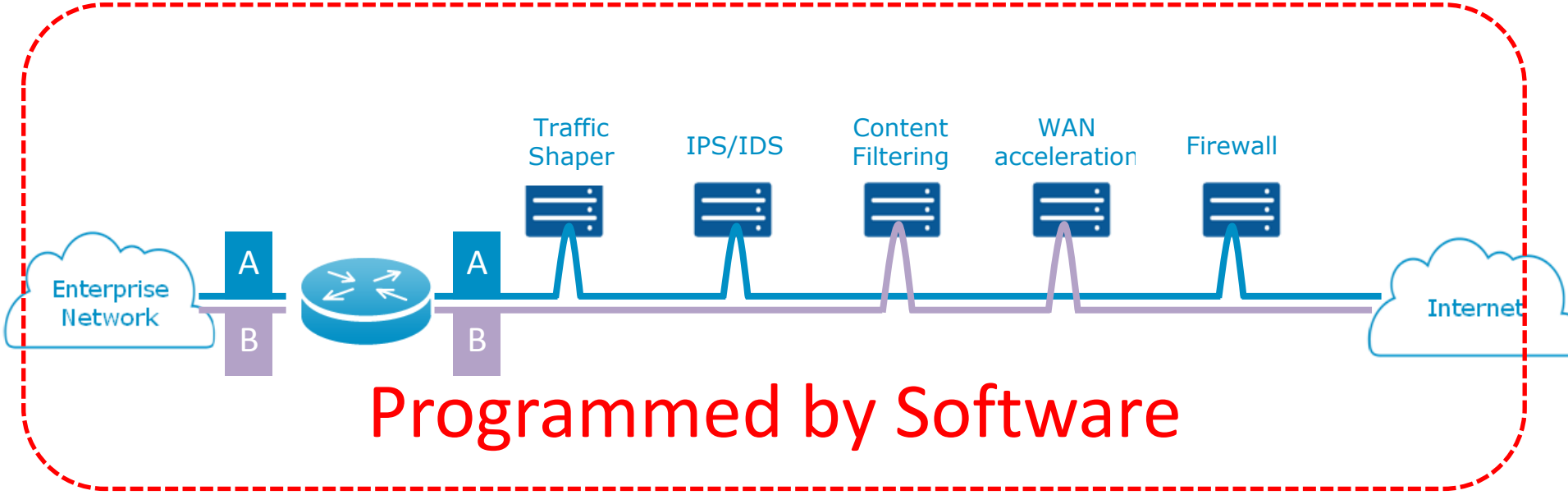


既存の手法でのService Chaining

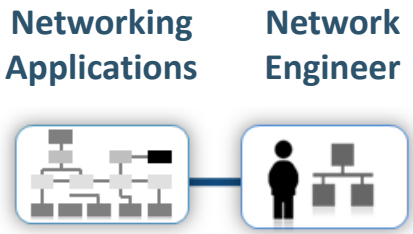


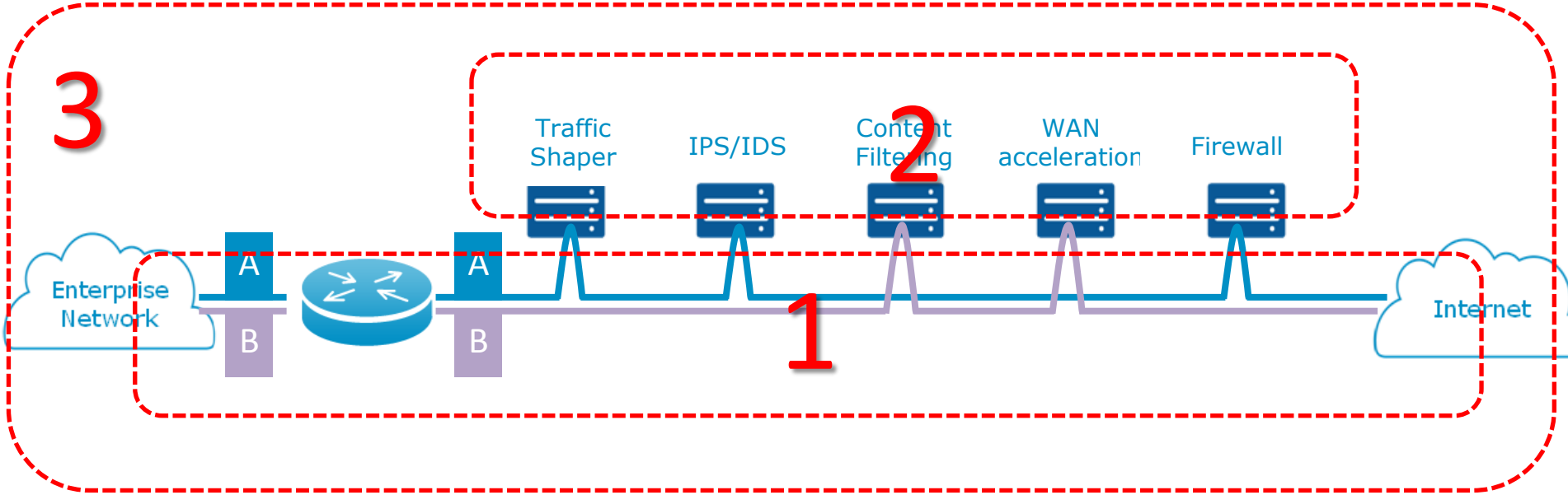
ダイナミックな Service Chaining



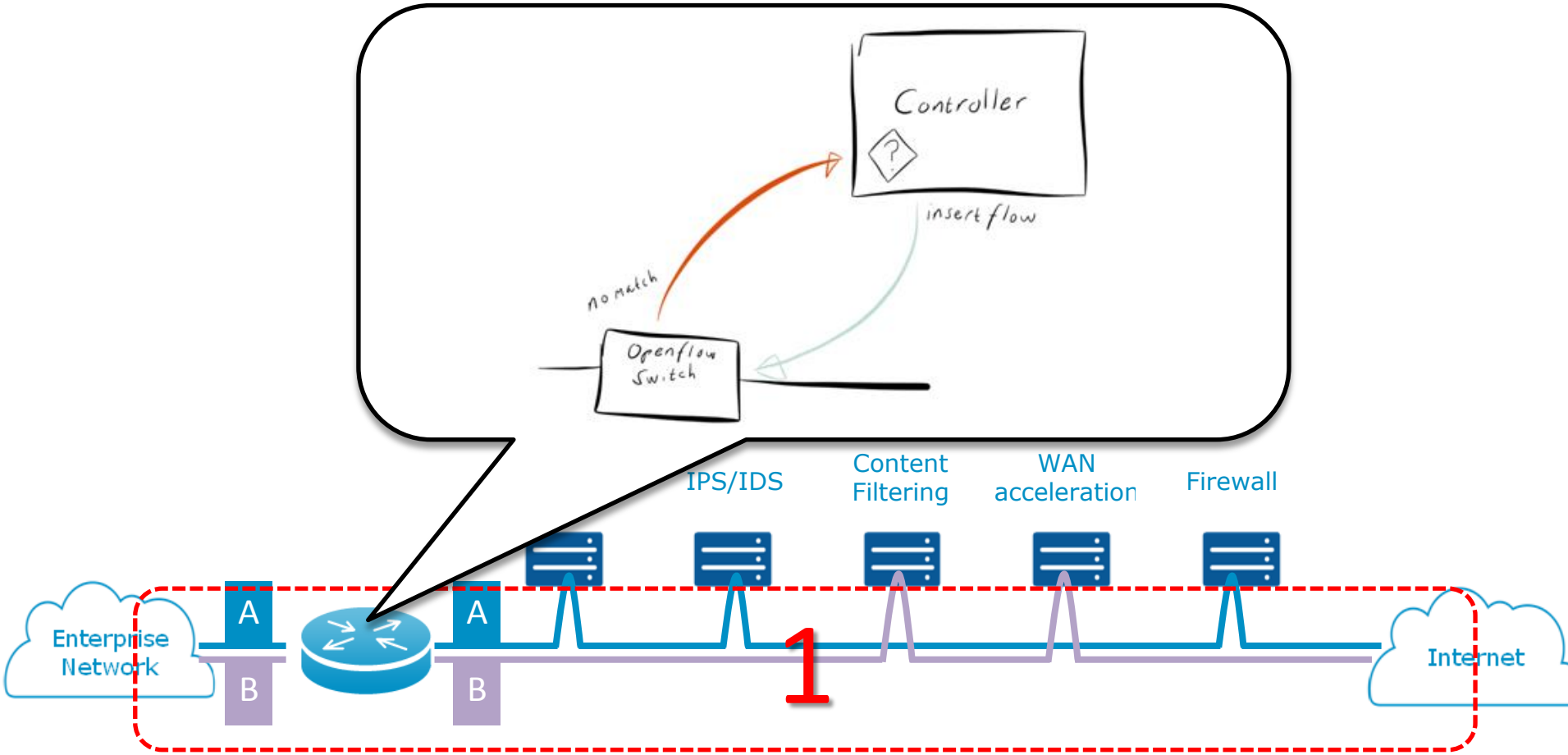


Service Chainingで要求されること

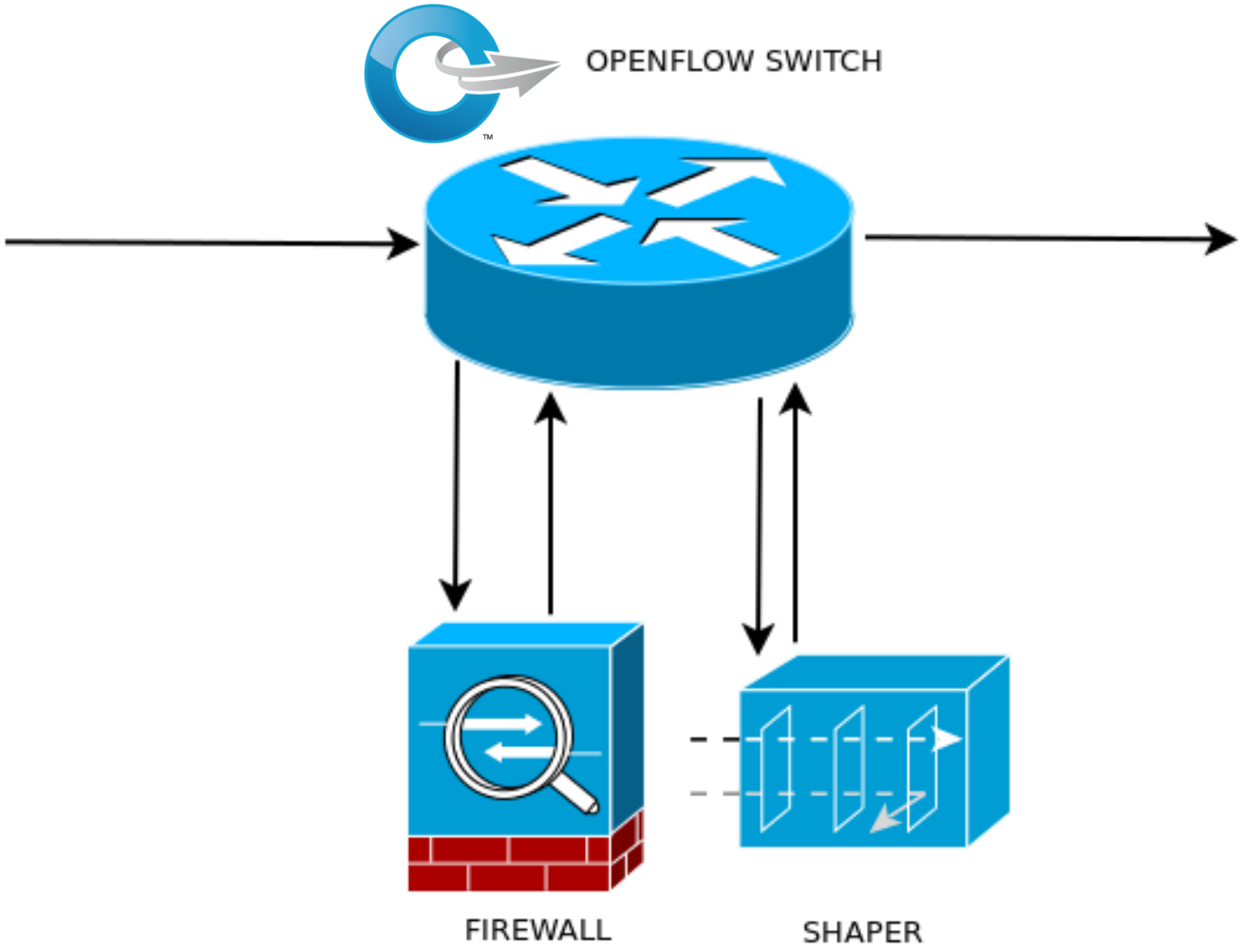




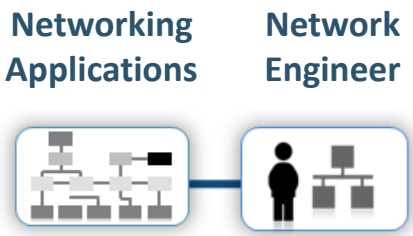
1への対応：OpenFlowの導入

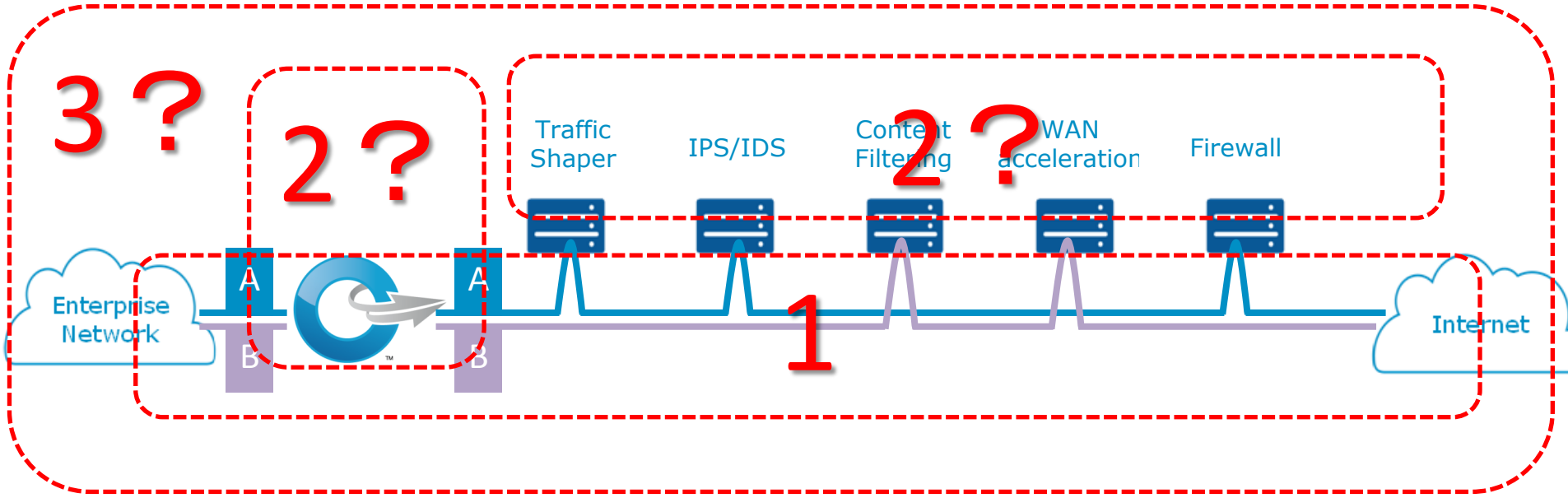


Openflowを組み合わせた構成



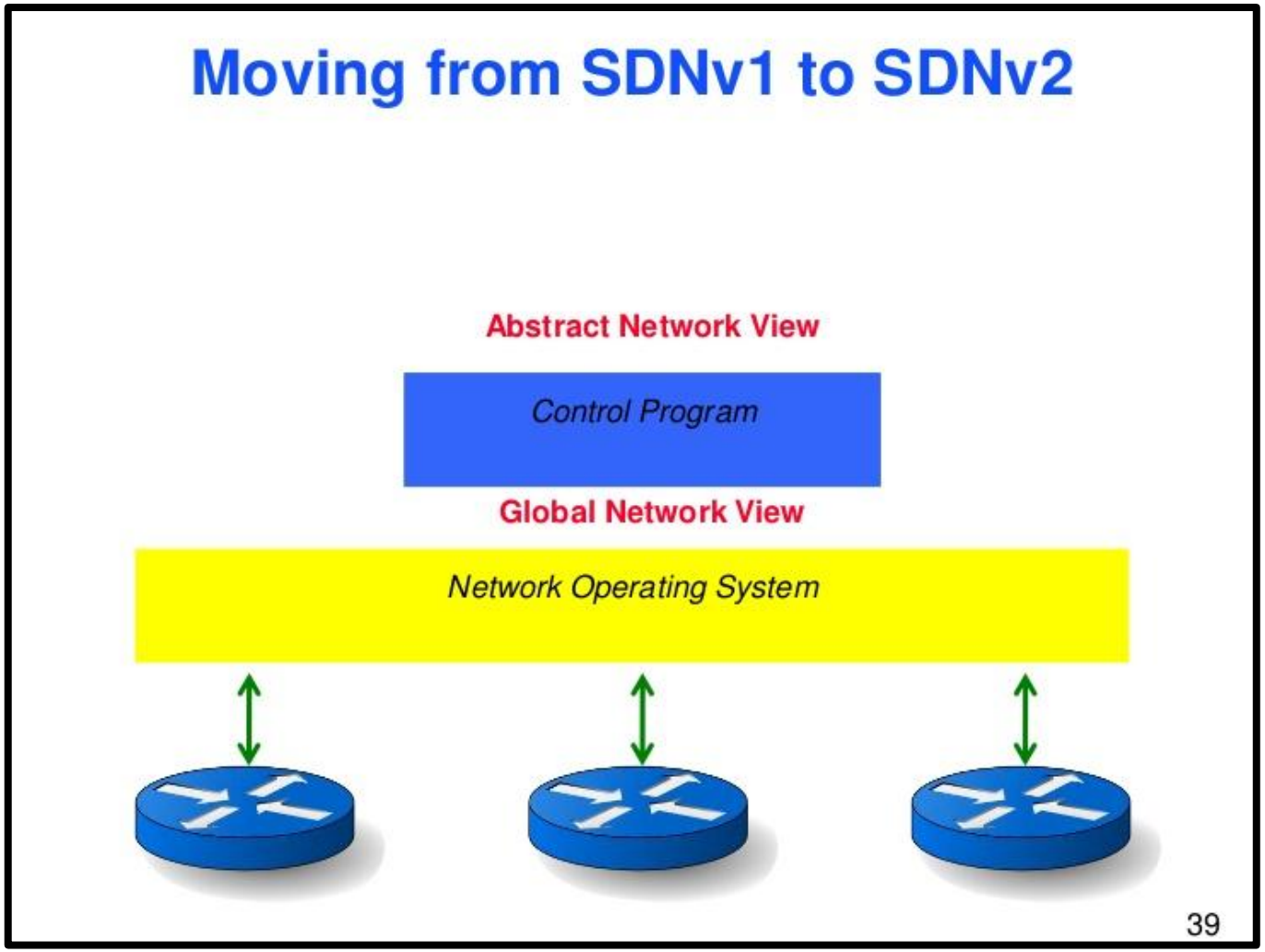
OpenFlowでのService Chaining





SDN Architecture

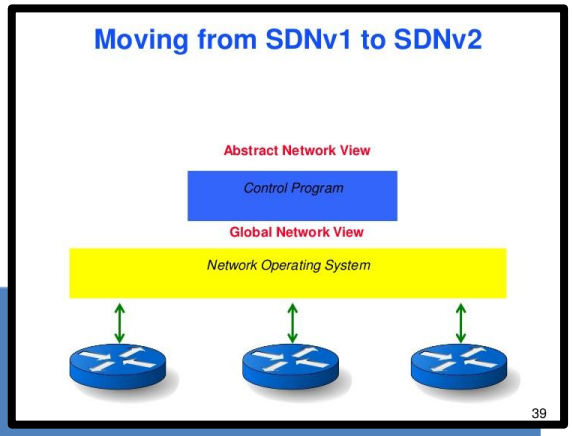
"The Future of Networking, and the Past of Protocols"



www.slideshare.net/martin_casado/sdn-abstractions

Service Chainingのモデル化



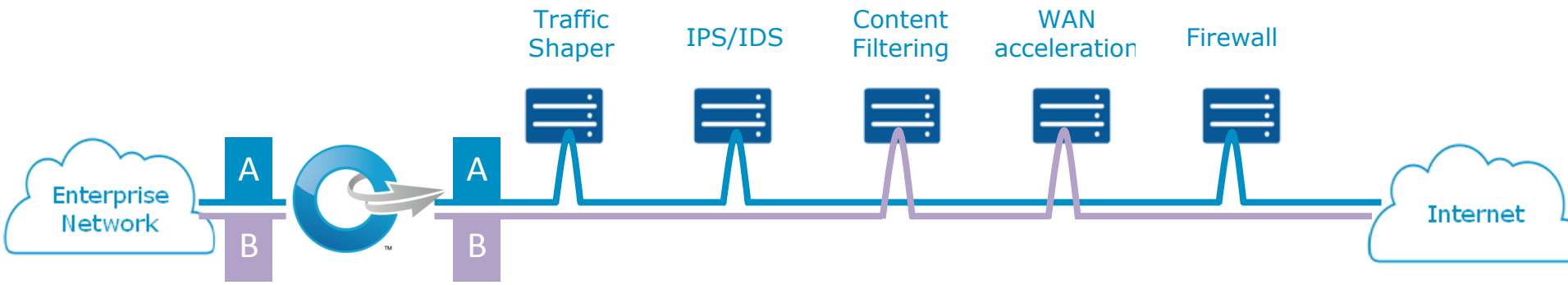


Service Model

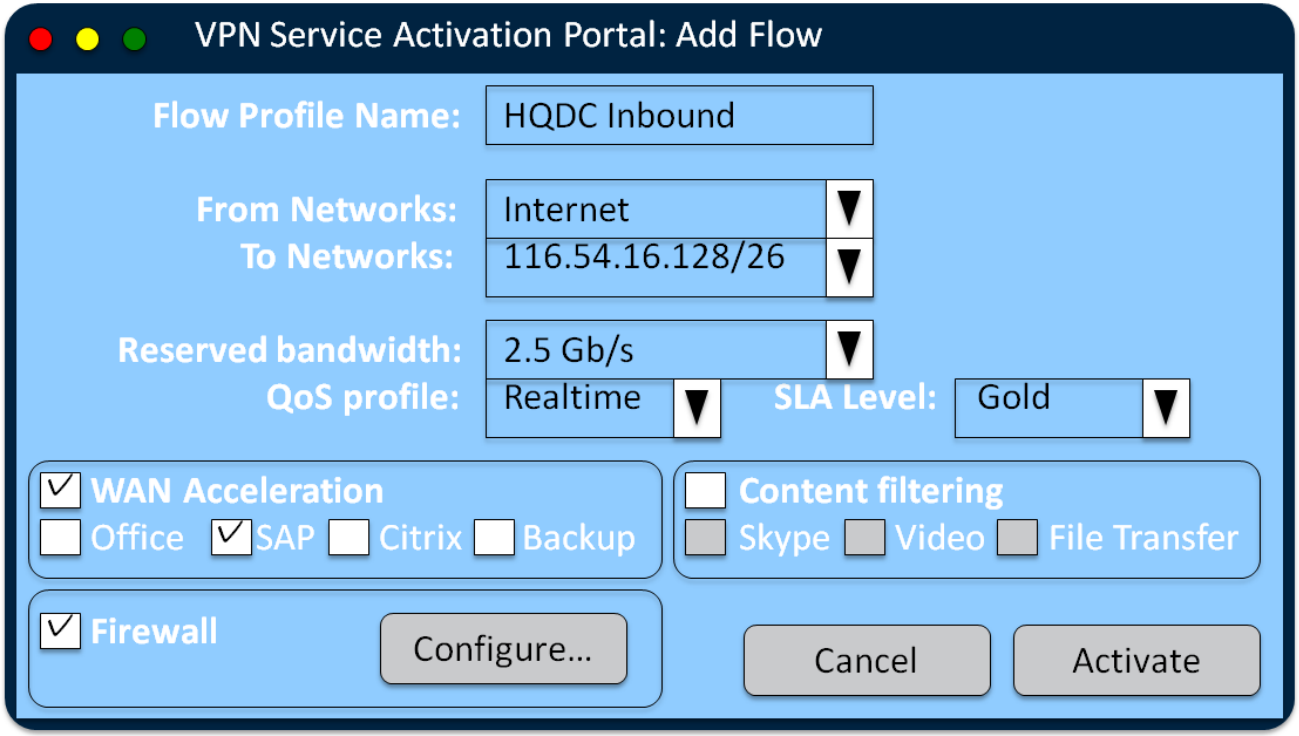
Service Chain

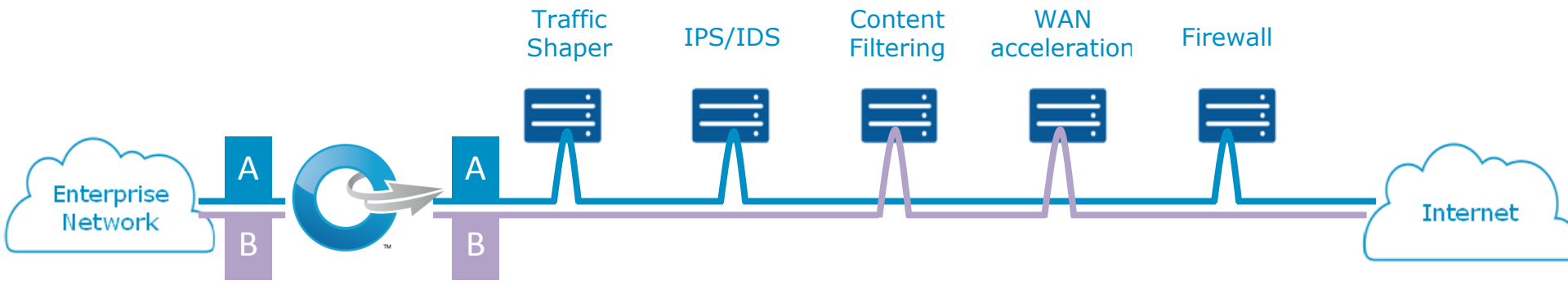
Traffic Steering | Traffic Shaper | IPS/IDS | Content Filtering | WAN acceleration | Firewall

Device Model

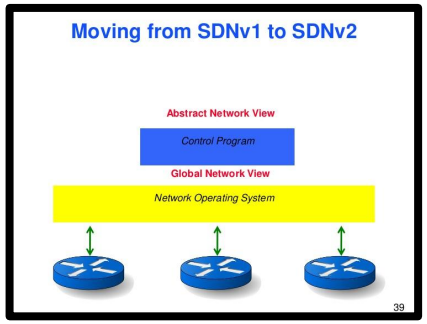


“Abstract Network View”





モデル間のマッピング



- VPN Service Model
  - Flow Profile Name: unique string
  - From Networks: [IP-address/mask]
  - To Networks: [IP-address/mask]
  - Reserved bandwidth: integer > 0

Service Application

- Traffic Steering Device Model
  - Port: unique integer 0..47
  - Src: IP-address/mask
  - Dst: IP-address/mask
  - Action: drop | output(N) | …
- Firewall Device Model
  - Src: IP-address/mask
  - Dst: IP-address/mask
  - Port: integer 1..65535
  - Action: drop | allow

ネットワーク全体への適用

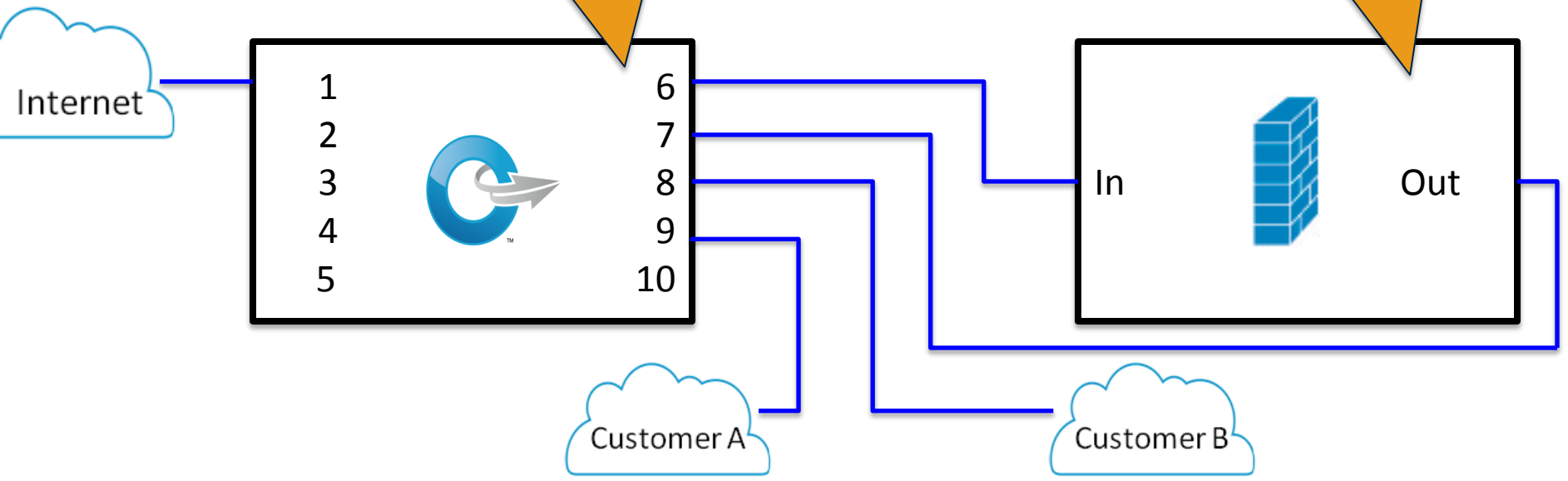
Network-wide Transaction

Rule #46: Port=1, Src=*, Dst=116.54.16.128/26, Action=output(6)

Rule #47: Port=7, Src=*, Dst=116.54.16.128/26, Action=output(8)

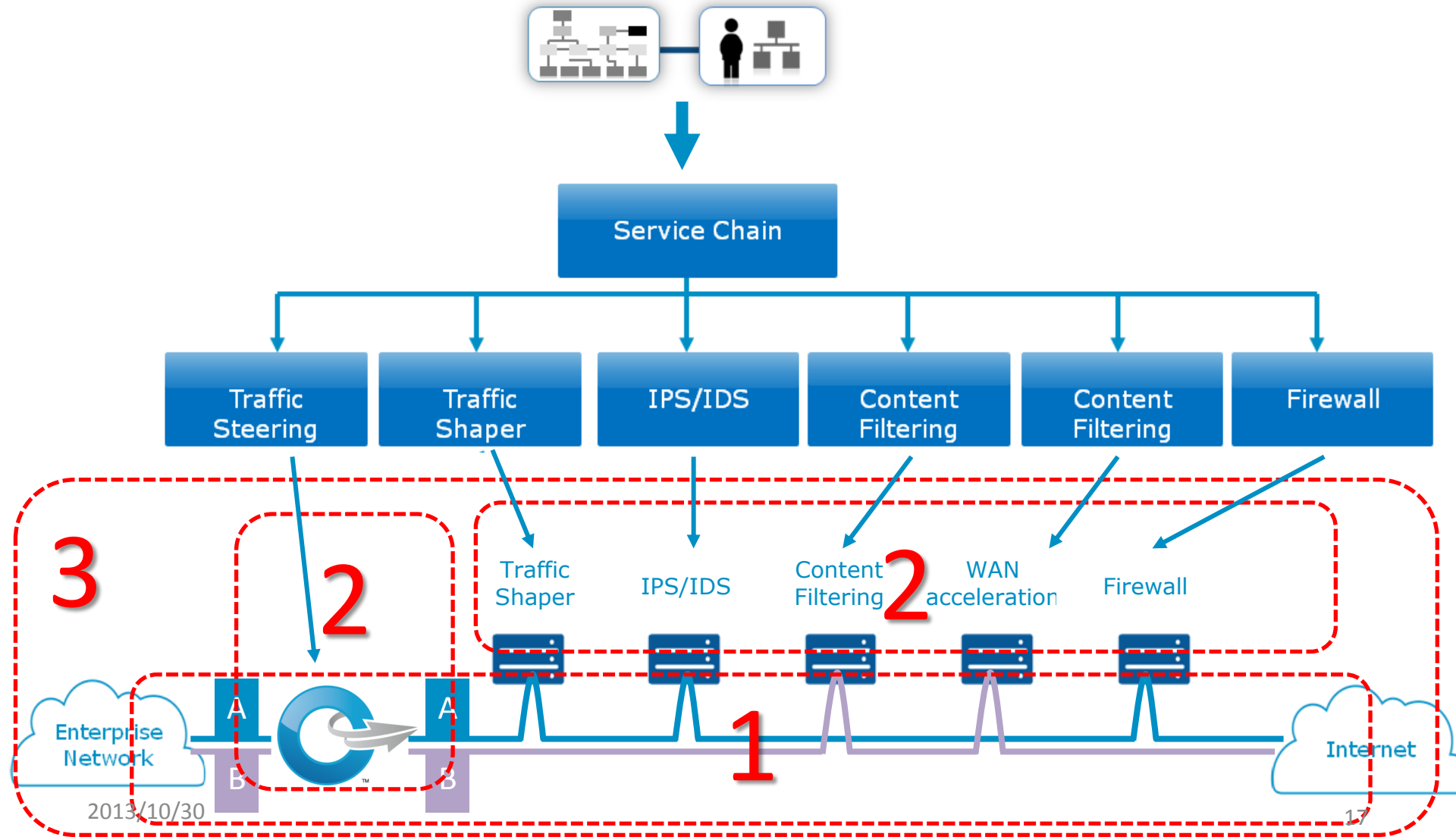
Rule #117: Src=*, Dst=116.54.16.128/26, Port=80, Action=allow

Rule #118: Src=*, Dst=116.54.16.128/26, Port=*, Action=drop

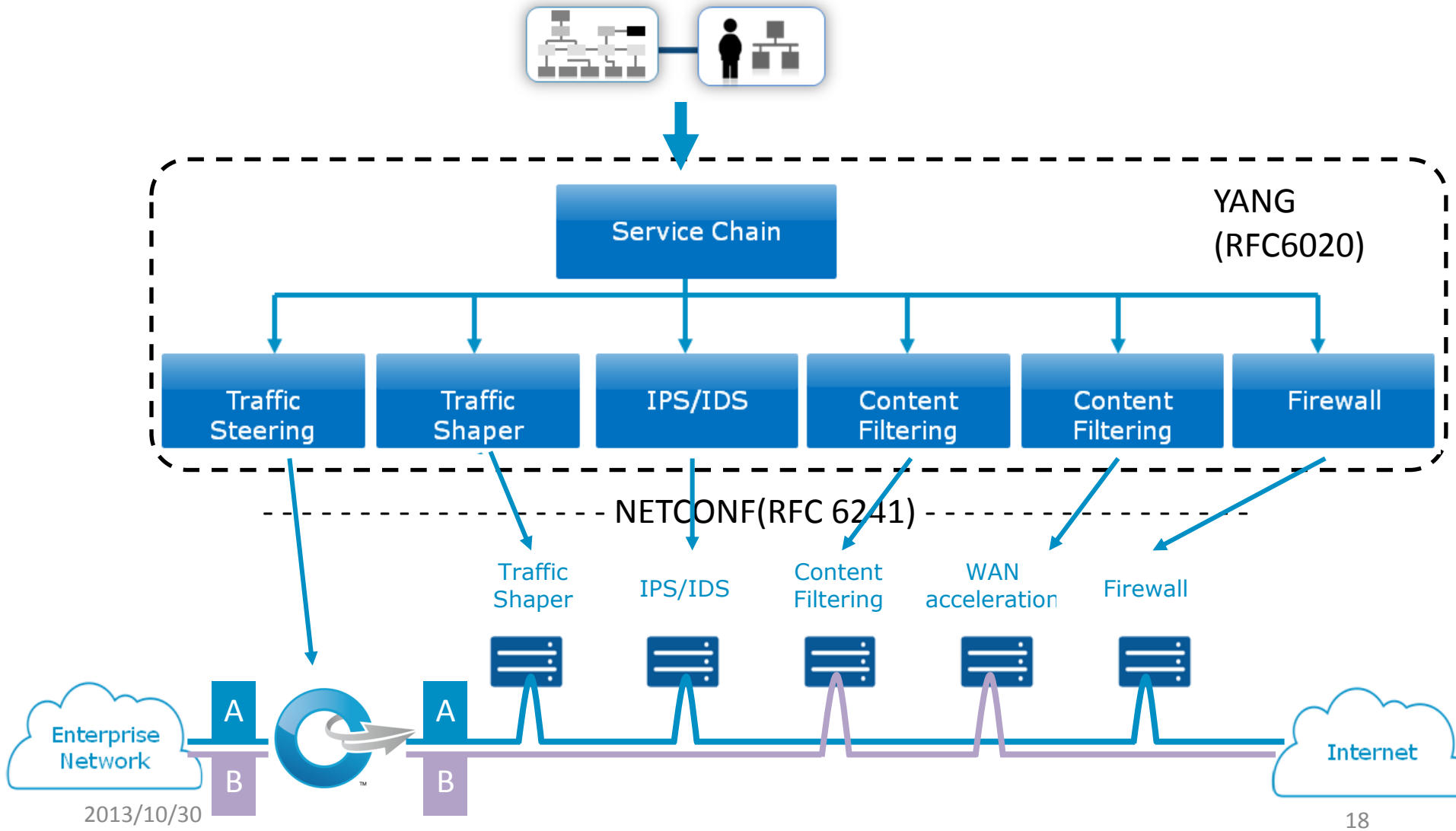


Demo

モデルベースでのService Chaning

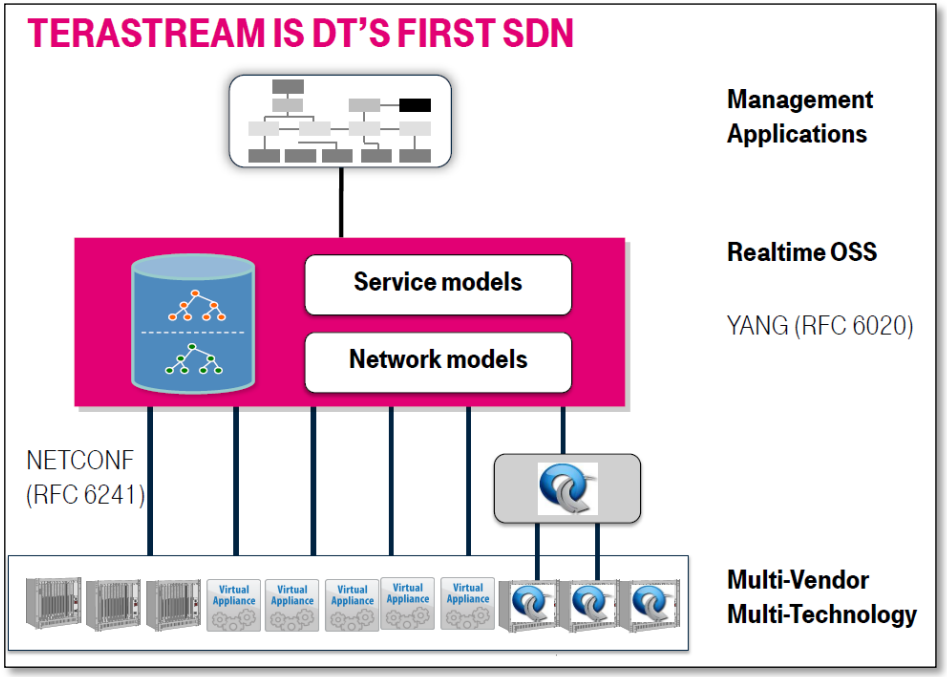


標準技術による汎用化


Service Chain
YANG
(RFC6020)
Traffic Steering
Traffic Shaper
IPS/IDS
Content Filtering
Content Filtering
Firewall
NETCONF(RFC 6241)
Traffic Shaper
IPS/IDS
Content Filtering
WAN acceleration
Firewall
Enterprise Network
A
B
A
B
Internet

ドイツテレコムの実績

- “A Realtime OSS-based SDN Approach”



- Deutsche Telekom: A Software-Defined Operator
  - http://www.lightreading.com/ethernet-ip/routers/deutsche-telekom-a-software-defined-operator/d/d-id/706099

ベンダー様でのNETCONF採用状況

- Alaxala
  - Ethernet switches
- BATM/Telco Systems
  - T-Metro 7224
- BigBand
  - MSP2800
- Brocade
  - NetIron XMR, CES, and CER
  - MLX Series
  - VDX
- Cisco
  - IOS 12.4(9)T and later
  - IOS XE 2.1 and later
- Edgeware
  - WTV-2X
- Ericsson
  - SEA 20
- H3C
  - S9500E Series Routing Switches
- Huawei
  - AR3200/2200 Enterprise Routers
- Juniper Networks
  - JUNOS 7.5 and later
- Nexor
  - Messaging Gateways
- RuggedCom
  - RX5000 and MX5000
- Sonus
  - NBS5200 Session Border Controller
- Taseon
  - TN 320
- Verivue
  - MDX 9020

Please Note that this list is work in progress and feedback on accuracy and completeness is strongly encouraged

標準化動向

IETF

- NETCONF WG
- NETMOD(YANG) WG

CableLabs

- YANG module for CCAP: next generation cable head-end systems

ONF
OPEN NETWORKING FOUNDATION

- OF-CONFIG YANG Modules

openstack
CLOUD SOFTWARE

- OpenStack Plugin

http://blog.ipspace.net/2013/10/openstack-quantum-neutron-plug-in-there.html

MEF
METRO ETHERNET FORUM

- YANG models for Service OAM PM and FM

Please Note that this list is work in progress and feedback on accuracy and completeness is strongly encouraged

Service Chainingで迅速なサービス実現、そして さらなる付加価値を！

